

NILESH ROY CHOWDHURY

Kolkata, India

+91-9007557560 | nileshroychowdhury6@gmail.com | GitHub | LinkedIn

Professional Summary

Computer Science undergraduate with strong foundations in Data Structures, Algorithms, Object-Oriented Programming, DBMS, and Software Development. Skilled in building responsive web applications and solving problems using Java, Python, C. Interested in AI-driven applications, with hands-on experience in NLP and web technologies.

Education

Institute of Engineering and Management

B.Tech in Computer Science and Engineering

Kolkata, India

July 2023 – Present

- Current CGPA: 8.27

The Central Modern School

Secondary Education (Class X)

Barasat, India

2021

The Central Modern School

Higher Secondary Education (Class XII)

Barasat, India

2023

Technical Skills

Programming Languages: Java, Python, C, C++, SQL, JavaScript

Web Technologies: HTML, CSS

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-Learn

Tools & Platforms: GitHub, MySQL, Jupyter Notebook, Google Colab, VS Code, IntelliJ IDEA, PyCharm

Core Computer Science Subjects: Data Structures & Algorithms, DBMS, Software Engineering

Soft Skills: Communication, Team Collaboration, Problem Solving, Leadership

Projects

Comparative Study of Transformer-Based Models for Bengali Hate Speech and Sarcasm Detection

Python, PyTorch, Hugging Face, LIME

- Developed a transformer-based NLP framework for classifying hate speech, sarcasm, and neutral text using multilingual Bengali social media datasets.
- Conducted a comparative evaluation of transformer models including XLM-R, MuRIL, and mBERT for low-resource Bengali language understanding and toxicity detection.
- Curated and preprocessed a multilingual Bengali dataset of 32,770+ samples using text normalization and sub-word tokenization techniques.
- Integrated explainable AI using LIME to improve model interpretability by identifying influential Bengali tokens that contributed to predictions.

Portfolio Website | HTML, Tailwind CSS, JavaScript

- Designed and developed a fully responsive portfolio website to showcase technical projects and skills.
- Implemented interactive UI components and smooth navigation using JavaScript and Tailwind CSS.
- Optimized website responsiveness across desktop, tablet, and mobile devices.
- Deployed the website using GitHub Pages for public access and project showcasing.

Certifications

Python for Data Science – NPTEL (IIT Madras)

Feb 2024

- Learned data analysis and visualization using Python, Pandas, NumPy, and Matplotlib.
- Successfully completed the proctored examination with a score of 69%.